Θέματα Στατιστικής στη γλώσσα R

- Η εμπειρική αθροιστική συνάρτηση κατανομής
- Γραφικός έλεγχος καλής προσαρμογής – qqplot

  Lab_3.pdf σελ.1-12
- Το κριτήριο των Kolmogorov-Smirnov (KS)
- Έλεγχοι για κανονικότητα
  - ✓ βασισμένος στο KS, έλεγχος του Lilliefors
  - ✓ έλεγχος των Shapiro–Wilk

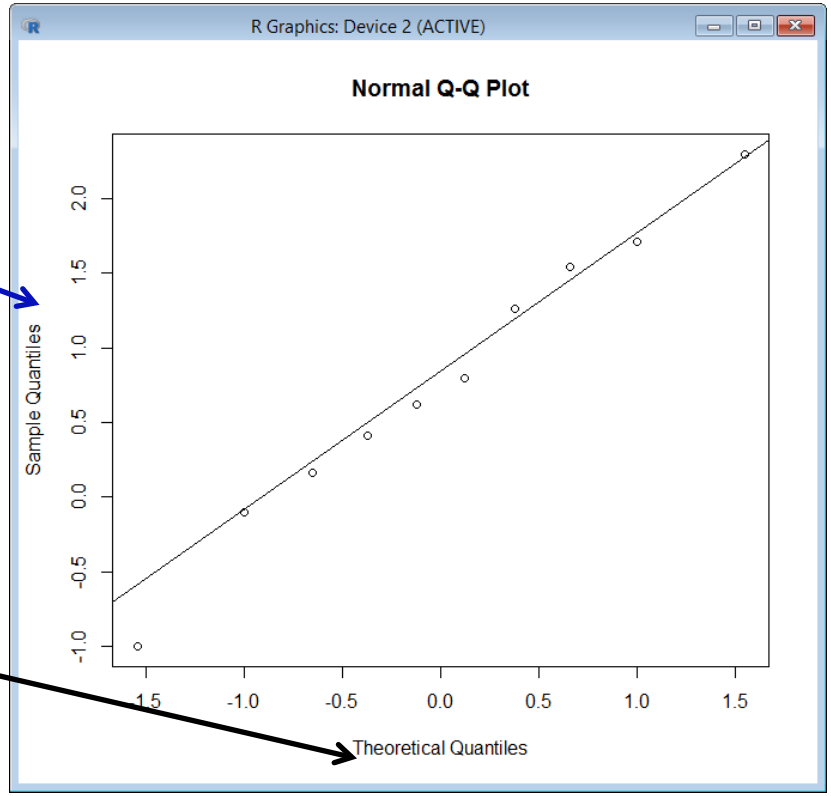
Οπτικός έλεγχος κανονικότητας-Κώδικας

Q-Q Plots - $\mathcal{N}(0,1)$

$$Q_1(p) = F^{-1}(p) = F^{-1}\left(\frac{i-0.5}{n}\right)$$

$x_{(i)}$	$\frac{i-0.5}{n}$	$Q_1(p)$
-1.00	0.05	-1.645
-0.10	0.15	-1.036
0.16	0.25	-0.674
0.41	0.35	-0.385
0.62	0.45	-0.125
0.80	0.55	0.125
1.26	0.65	0.385
1.54	0.75	0.674
1.71	0.85	1.036
2.30	0.95	1.645

```
> x<-c(-1,-0.1,0.16,0.41,0.62,
+ 0.80,1.26,1.54,1.71,2.30)
> qqnorm(x)
Waiting to confirm page change...
> qqline(x)
```



Οπτικός έλεγχος κανονικότητας-Κώδικας

Lab_3.pdf σελ.7-8, 11-12

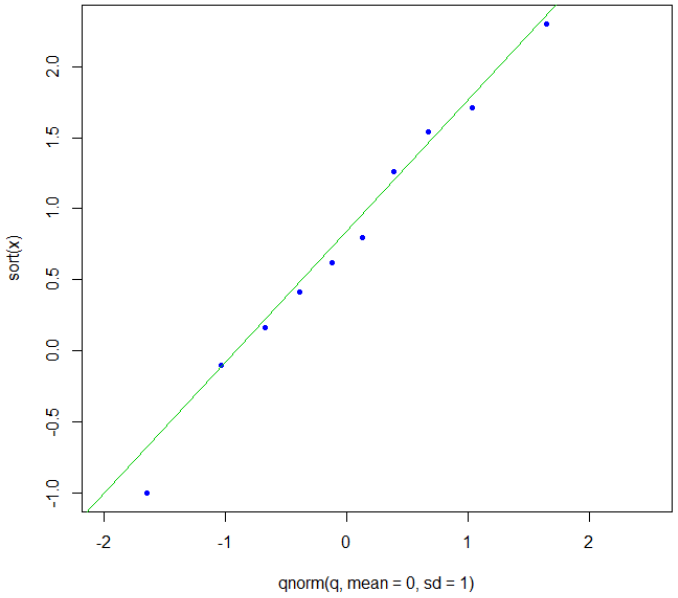
```
> n<-length(x)
> q<-seq((1-0.5)/n,(n-0.5)/n,1/n)
> # q<-seq(from=(1-0.5)/n,by=1/n,length.out=n)
> round(qnorm(q,mean=0,sd=1),3)
 [1] -1.645 -1.036 -0.674 -0.385 -0.126  0.126  0.385
 [8]  0.674  1.036  1.645
> cbind(sort(x),q,round(qnorm(q,mean=0,sd=1),3))
                 q
 [1,] -1.00 0.05 -1.645
 [2,] -0.10 0.15 -1.036
 [3,]  0.16 0.25 -0.674
 [4,]  0.41 0.35 -0.385
 [5,]  0.62 0.45 -0.126
 [6,]  0.80 0.55  0.126
 [7,]  1.26 0.65  0.385
 [8,]  1.54 0.75  0.674
 [9,]  1.71 0.85  1.036
[10,]  2.30 0.95  1.645
```

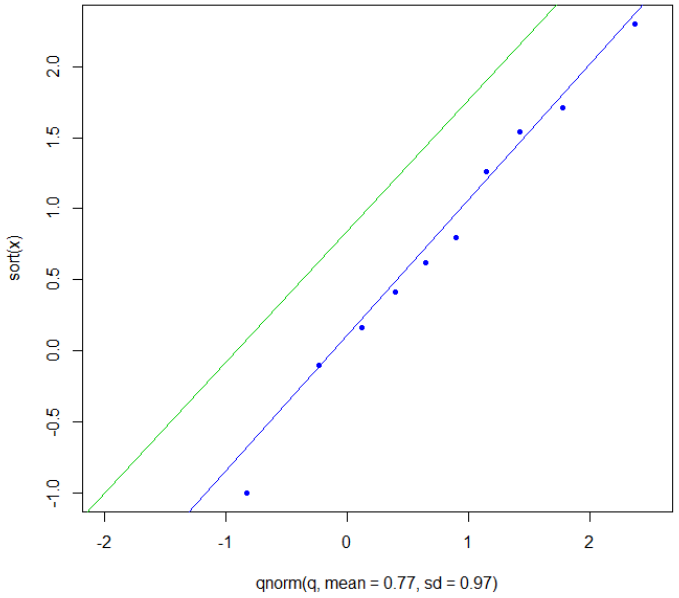
$$Q_1(p) = F^{-1}(p) = F^{-1}\left(\frac{i-0.5}{n}\right)$$

$x_{(i)}$	$\frac{i-0.5}{n}$	$Q_1(p)$
-1.00	0.05	-1.645
-0.10	0.15	-1.036
0.16	0.25	-0.674
0.41	0.35	-0.385
0.62	0.45	-0.125
0.80	0.55	0.125
1.26	0.65	0.385
1.54	0.75	0.674
1.71	0.85	1.036
2.30	0.95	1.645

Οπτικός έλεγχος καλής προσαρμογής σε κατανομή -Κώδικας

```
> par(mfrow=c(1,2))
> plot(qnorm(q,mean=0,sd=1),sort(x),pch=20,col="blue",xlim=c(-2,2.5))
Waiting to confirm page change...
> qqline(distribution = function(p) qnorm(p,mean=0,sd=1),x,
+        prob = c(0.25, 0.75), col = 3)
>
> plot(qnorm(q,mean=0.77,sd=0.97),sort(x),pch=20,col="blue",xlim=c(-2,2.5))
> qqline(distribution = function(p) qnorm(p,mean=0.77,sd=0.97),x,
+        prob = c(0.25, 0.75), col ="blue")
> qqline(distribution = function(p) qnorm(p,mean=0,sd=1), x,
+        prob = c(0.25, 0.75), col = 3)
```



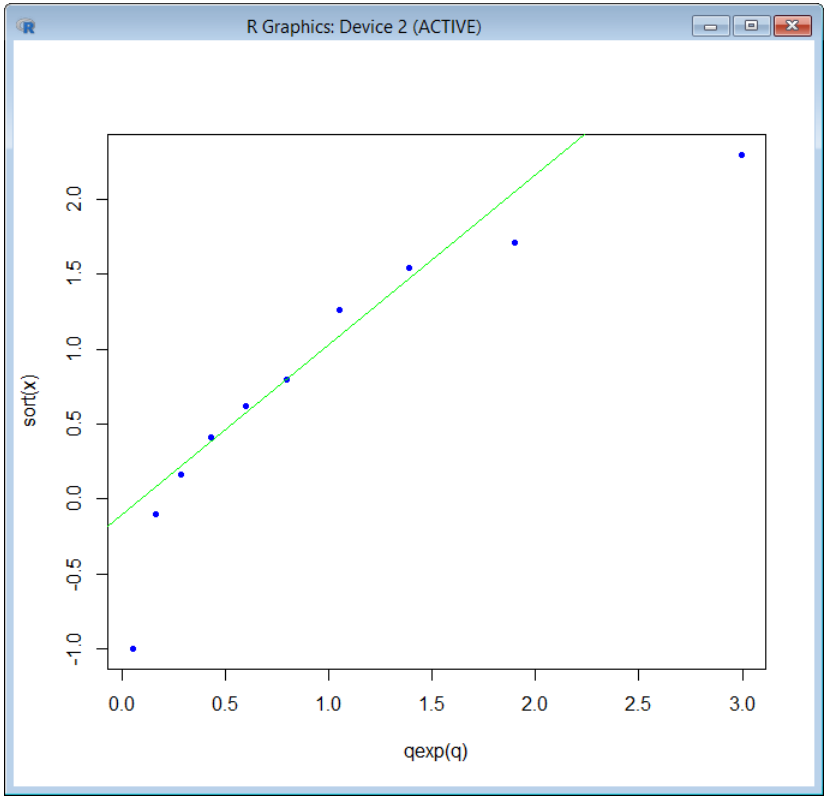


Οπτικός έλεγχος καλής προσαρμογής σε Εκθετική(1) -Κώδικας

γίνεται για οποιαδήποτε κατανομή αντικαθιστώντας την

$\Phi^{-1}((i-0.5)/n)$=qnorm((i-0.5)/n)

από την αντίστροφη συνάρτηση της αντίστοιχης αθροιστικής




```
> plot(qexp(q),sort(x),pch=20,col="blue")
Waiting to confirm page change...
> qqline(x, distribution = function(p) qexp(p, rate=1),
+        prob = c(0.25, 0.75), col = 3)
```

Το κριτήριο των Kolmogorov-Smirnov (KS)

Έλεγχος Υπόθεσης (για ένα δείγμα)
εάν προέρχεται από κατανομή (με συγκεκριμένες παραμέτρους)

$H_0: F(x)=F_0(x)$ για κάθε $x \in \mathbb{R}$ $H_0: F(x)=F_0(x)$ $H_0: F(x)=F_0(x)$
$H_1: F(x)\neq F_0(x)$ $H_1: F(x)>F_0(x)$ $H_1: F(x)<F_0(x)$

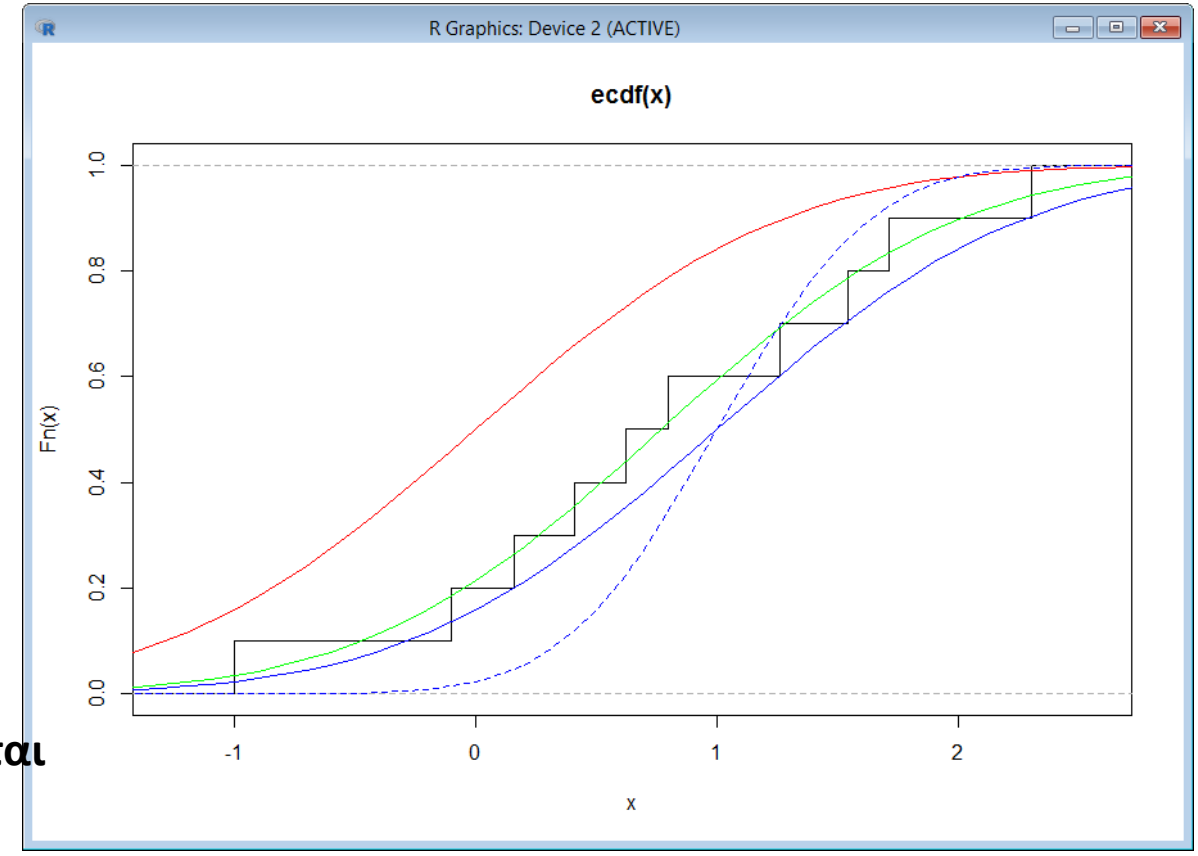
$F_0 \equiv N(0,1)$

$F_0 \equiv N(1,1)$
$F_0 \equiv N(1,0.5^2)$ ---

~~$F_0 \equiv N(M_x, S_x^2)$~~

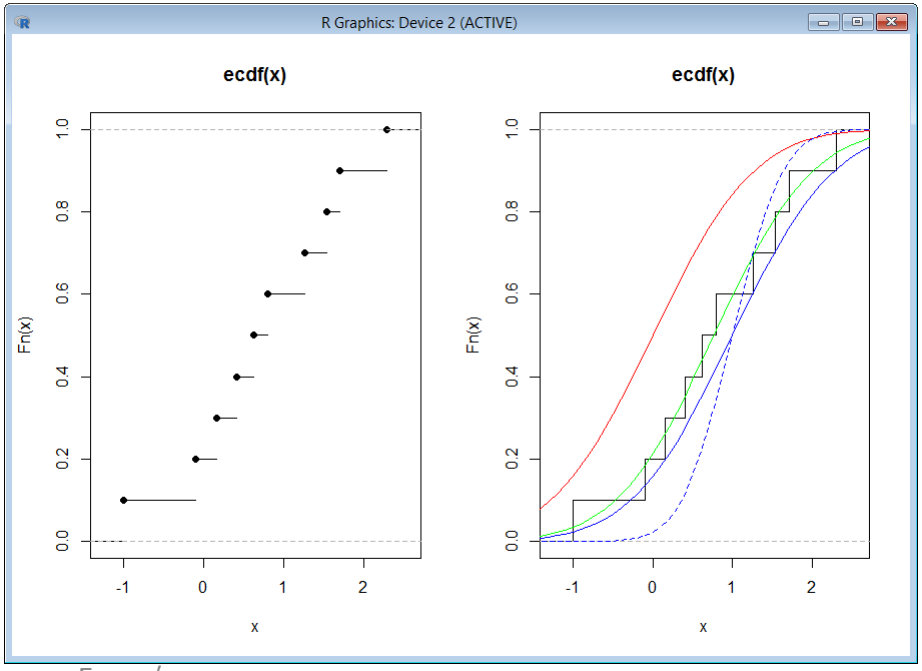
όπου
$M_x = \text{mean}(x) = 0.77$
$S_x^2 = \text{var}(x) = 0.97^2$

Δεν επιτρέπεται,
οι παράμετροι εξαρτώνται
από το δείγμα



Το κριτήριο των Kolmogorov-Smirnov (KS)

```
> sort(x)
 [1] -1.00 -0.10  0.16  0.41  0.62  0.80  1.26  1.54  1.71  2.30
> Fx<-ecdf(x)
> plot(Fx)
Waiting to confirm page change...
> plot(Fx, do.points=FALSE, verticals=TRUE)
> Mx<-mean(x)
> Sx<-sd(x)
> curve(pnorm(x,mean=0,sd=1), add=TRUE, xlim=c(-5,5),col="red")
> curve(pnorm(x,mean=1,sd=1), add=TRUE, xlim=c(-5,5),col="blue")
> curve(pnorm(x,mean=1,sd=0.5), add=TRUE, xlim=c(-5,5),col="blue",lty=2)
> curve(pnorm(x,mean=Mx,sd=Sx), add=TRUE, xlim=c(-5,5),col="green")
```

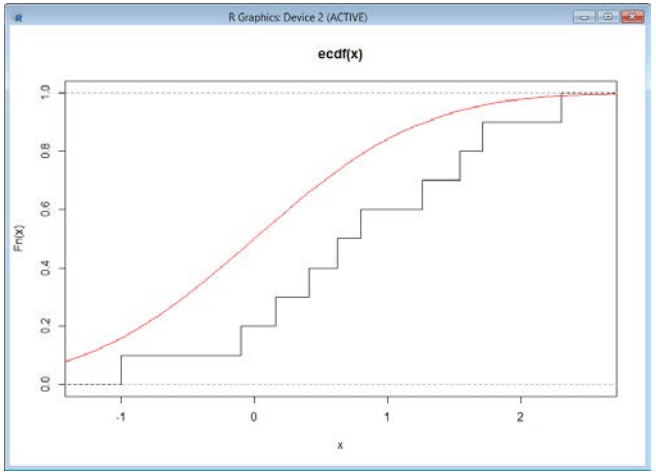


Σύγκριση της εμπειρικής α.σ.κ. του δείγματος με την α.σ.κ. που αντιστοιχεί στην $\mathbf{F_0}$

Βλέπε Lab_3.pdf σελ. 3-6, 9

Το κριτήριο των Kolmogorov-Smirnov (KS)

$F_0 \equiv N(0,1)$




```
> ks.test(x,"pnorm")

        One-sample Kolmogorov-Smirnov test

data:  x
D = 0.36356, p-value = 0.1085
alternative hypothesis: two-sided

> ks.test(x,"pnorm",mean=0,sd=1,alternative ="greater")

        One-sample Kolmogorov-Smirnov test

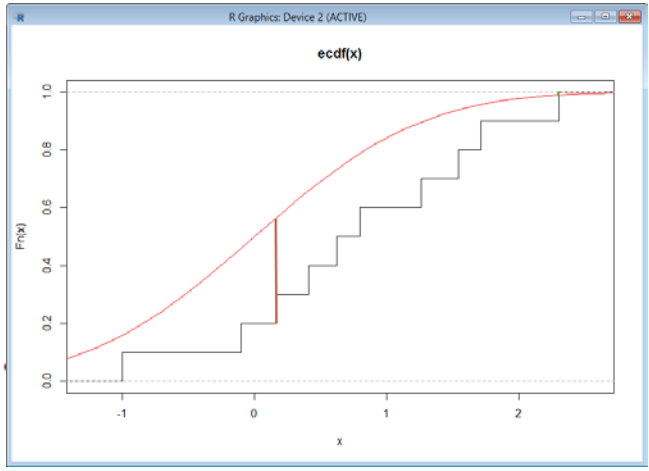
data:  x
D^+ = 0.010724, p-value = 0.9882
alternative hypothesis: the CDF of x lies above the null hypothesis

> ks.test(x,"pnorm",mean=0,sd=1,alternative ="less")

        One-sample Kolmogorov-Smirnov test

data:  x
D^- = 0.36356, p-value = 0.05425
alternative hypothesis: the CDF of x lies below the null hypothesis
```

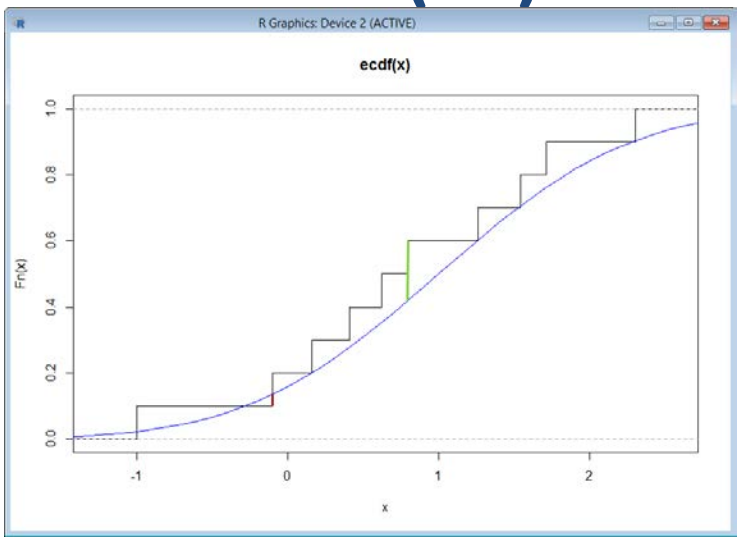
Το κριτήριο των Kolmogorov-Smirnov (KS)




```
> n<-length(x)
> order.x<-sort(x)
> ecdf(x)(order.x)
 [1] 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.
> Fn<-ecdf(x)(order.x)
> Fn
 [1] 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
> F0<-pnorm(order.x,mean=0,sd=1)
> cbind(order.x,Fn,F0,Fn-F0,F0-c(0,Fn[-n]))
      order.x  Fn        F0
 [1,]   -1.00 0.1 0.1586553 -0.05865525 0.15865525
 [2,]   -0.10 0.2 0.4601722 -0.26017216 0.36017216
 [3,]    0.16 0.3 0.5635595 -0.26355946 0.36355946
 [4,]    0.41 0.4 0.6590970 -0.25909703 0.35909703
 [5,]    0.62 0.5 0.7323711 -0.23237111 0.33237111
 [6,]    0.80 0.6 0.7881446 -0.18814460 0.28814460
 [7,]    1.26 0.7 0.8961653 -0.19616532 0.29616532
 [8,]    1.54 0.8 0.9382198 -0.13821982 0.23821982
 [9,]    1.71 0.9 0.9563671 -0.05636706 0.15636706
[10,]    2.30 1.0 0.9892759  0.01072411 0.08927589
```

Το κριτήριο των Kolmogorov-Smirnov (KS)

$F_0 \equiv N(1,1)$




```
> ks.test(x,"pnorm",mean=1,sd=1)

        One-sample Kolmogorov-Smirnov test

data:  x
D = 0.17926, p-value = 0.8506
alternative hypothesis: two-sided

> ks.test(x,"pnorm",mean=1,sd=1,alternative ="greater")

        One-sample Kolmogorov-Smirnov test

data:  x
D^+ = 0.17926, p-value = 0.4709
alternative hypothesis: the CDF of x lies above the null hypothesis

> ks.test(x,"pnorm",mean=1,sd=1,alternative ="less")

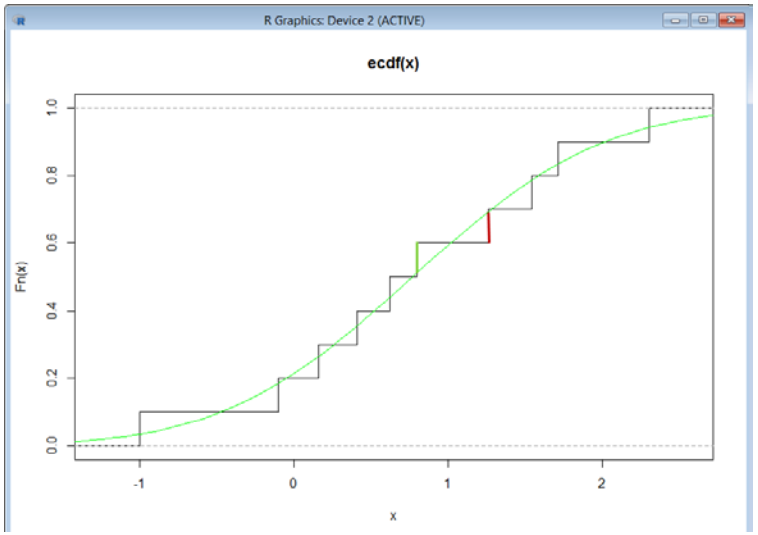
        One-sample Kolmogorov-Smirnov test

data:  x
D^- = 0.035666, p-value = 0.9511
alternative hypothesis: the CDF of x lies below the null hypothesis
```

Το κριτήριο των Kolmogorov-Smirnov (KS)

$F_0 \equiv N(M_x, S_x^2)$ όπου M_x = mean(x)= 0.77

S_x^2= var(x)= 0.97^2




```
> ks.test(x,"pnorm",mean=mean(x),sd=sd(x))

        One-sample Kolmogorov-Smirnov test

data:  x
D = 0.09323, p-value = 0.9999
alternative hypothesis: two-sided

> ks.test(x,"pnorm",mean=mean(x),sd=sd(x),alternative ="greater")

        One-sample Kolmogorov-Smirnov test

data:  x
D^+ = 0.087667, p-value = 0.8132
alternative hypothesis: the CDF of x lies above the null hypothesis

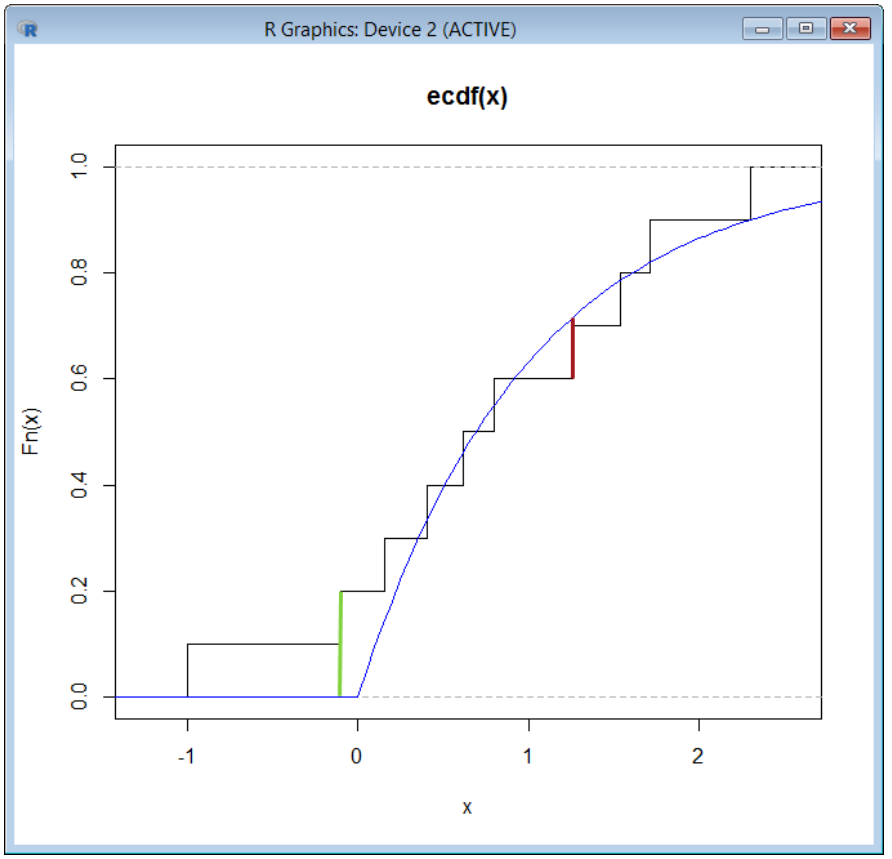
> ks.test(x,"pnorm",mean=mean(x),sd=sd(x),alternative ="less")

        One-sample Kolmogorov-Smirnov test

data:  x
D^- = 0.09323, p-value = 0.7921
alternative hypothesis: the CDF of x lies below the null hypothesis
```

Το κριτήριο των Kolmogorov-Smirnov (KS)

$F_0 \equiv Exp(1)$




```
> ks.test(x,"pexp",rate=1)

        One-sample Kolmogorov-Smirnov test

data:  x
D = 0.2, p-value = 0.7487
alternative hypothesis: two-sided
```

Έλεγχος για κανονικότητα

✓ έλεγχος των Shapiro–Wilk

εάν προέρχεται από κανονική κατανομή (με οποιεσδήποτε παραμέτρους)

H_0: $F(x)=N(\mu,\sigma^2)$ για κάποια $\mu \in \mathbb{R}, \sigma^2>0$

H_1: $F(x)\neq N(\mu,\sigma^2)$ για όλα τα δυνατά ζεύγη (μ, σ^2)

```
> shapiro.test(x)

        Shapiro-Wilk normality test

data:  x
W = 0.98996, p-value = 0.9968

> y<-scan()
1: 0.12 1.01 0.17 1.77 0.27 0.98 0.11 0.12 0.01 0.34
11:
Read 10 items
> shapiro.test(y)

        Shapiro-Wilk normality test

data:  y
W = 0.7789, p-value = 0.008028
```

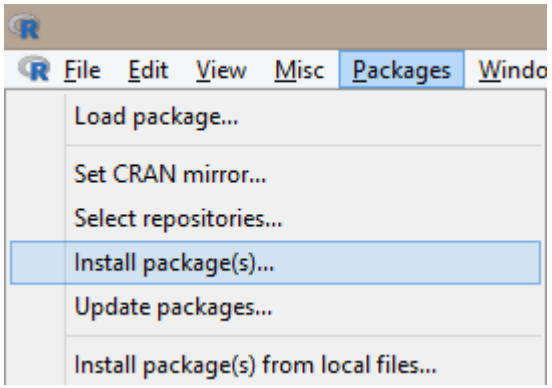
(δείγμα από Exp(1))

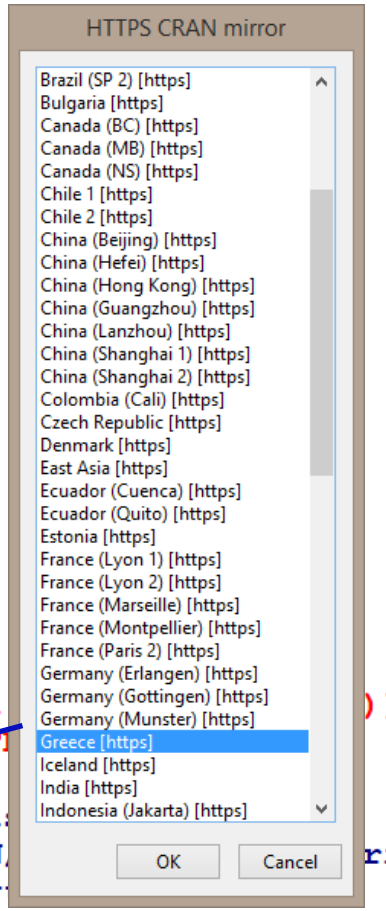
Έλεγχος για κανονικότητα

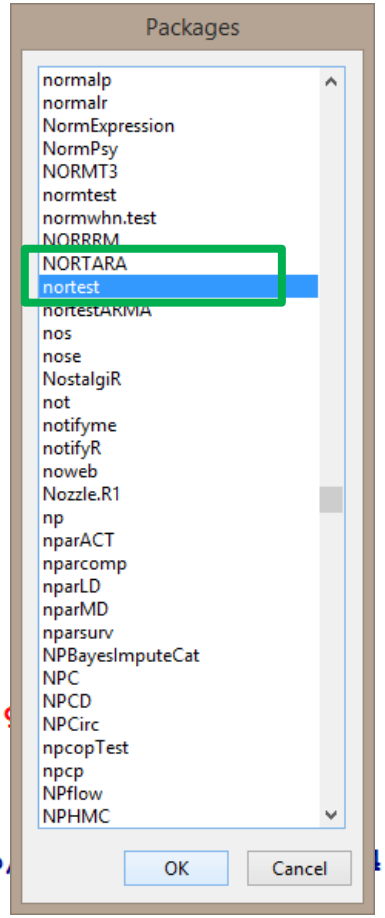
✓ βασισμένος στο KS, έλεγχος του Lilliefors

```
> lillie.test(x)
Error: could not find function "lillie.test"
```

Comprehensive **R** **A**rchive **N**etwork








```
> local({pkg <- select.list(sort(.packages(all
+ if(nchar(pkg)) library(pkg, character.only=T
> utils:::menuInstallPkgs()
--- Please select a CRAN mirror for use in this
trying URL 'https://ftp.cc.uoc.gr/mirrors/CRAN/
Content type 'application/zip' length 30329 by
downloaded 29 KB

package 'nortest' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
        C:\Users\Violeta Piperigkou\AppData\Local\Temp\Rtmp2hw4NT\downloaded_packages
```

Έλεγχος για κανονικότητα

- ✓ βασισμένος στο KS, έλεγχος του Lilliefors

https://cran.r-project.org/web/packages/nortest/nortest.pdf

```
> library(nortest)
> lillie.test(x)

        Lilliefors (Kolmogorov-Smirnov) normality test

data:  x
D = 0.09323, p-value = 0.999

> ks.test(x,"pnorm",mean=mean(x),sd=sd(x))

        One-sample Kolmogorov-Smirnov test

data:  x
D = 0.09323, p-value = 0.9999
alternative hypothesis: two-sided
```

έτυχε

Έλεγχος για κανονικότητα

- ✓ βασισμένος στο KS, έλεγχος του Lilliefors

```
> y
 [1] 0.12 1.01 0.17 1.77 0.27 0.98 0.11 0.12 0.01 0.34

> lillie.test(y)

        Lilliefors (Kolmogorov-Smirnov) normality test

data:  y
D = 0.30299, p-value = 0.009772

> ks.test(y,"pnorm",mean=mean(y),sd=sd(y))

        One-sample Kolmogorov-Smirnov test

data:  y
D = 0.30299, p-value = 0.3176
alternative hypothesis: two-sided

Warning message:
In ks.test(y, "pnorm", mean = mean(y), sd = sd(y)) :
  ties should not be present for the Kolmogorov-Smirnov test
```

Έλεγχος για κανονικότητα

✓ βασισμένος στο KS, έλεγχος του Lilliefors

```
> lillie.test(x)

        Lilliefors (Kolmogorov-Smi:

data:  x
D = 0.09323, p-value = 0.999
```

SPSS Kolmogorov-Smirnov

Tests of Normality

	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
X	,093	10	,200*	,990	10	,997

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

```
>   ks.gof(x)
```

S-Plus ks.gof

```
        One sample Kolmogorov-Smirnov Test of Composite Normality

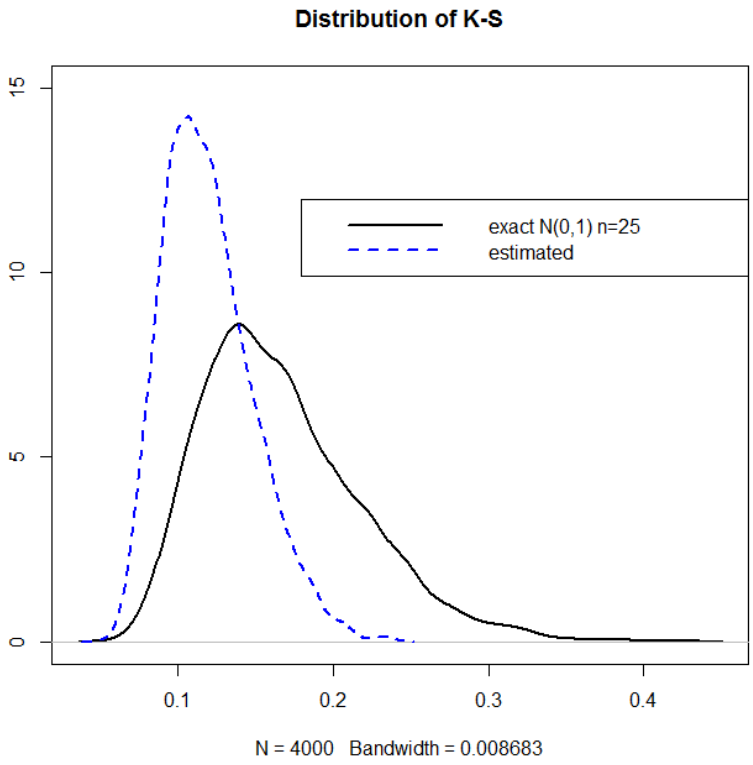
data:  x
ks = 0.09323, p-value = 0.5
alternative hypothesis: True cdf is not the normal distn. with estimated parameters
sample estimates:
             mean of x standard deviation of x
             0.7700000                0.9702462

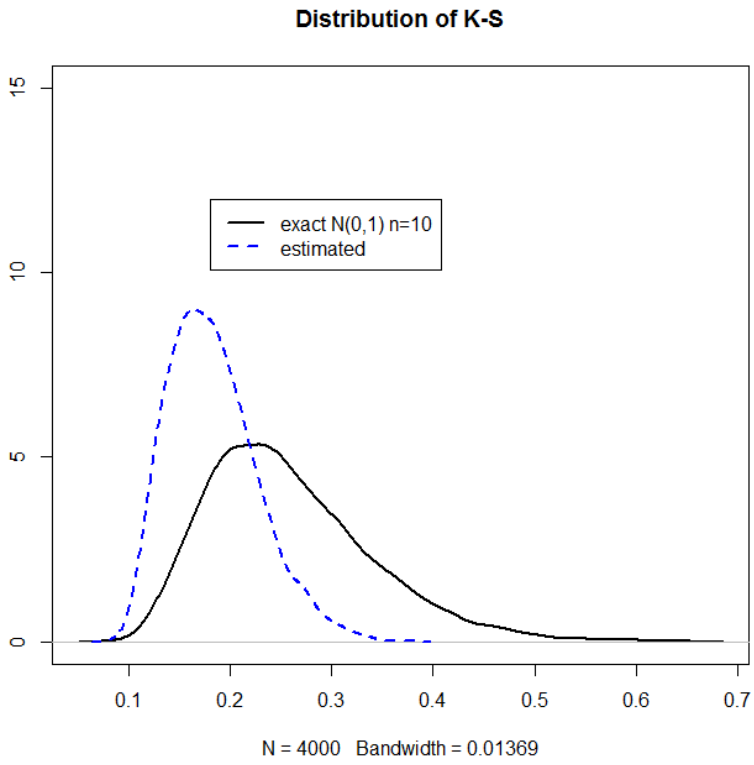
Warning message:
In dall.wilk(test, nx) :
  The Dallal-Wilkinson approximation, used to calculate
     the p-value in testing composite normality,
     is most accurat
e for p-values <= 0.10 .
   The calculated p-value is 1.775 and so is set to 0.5 .
```

Έλεγχος για κανονικότητα

- ✓ έλεγχος του Lilliefors
- ✓ το κριτήριο των Kolmogorov-Smirnov

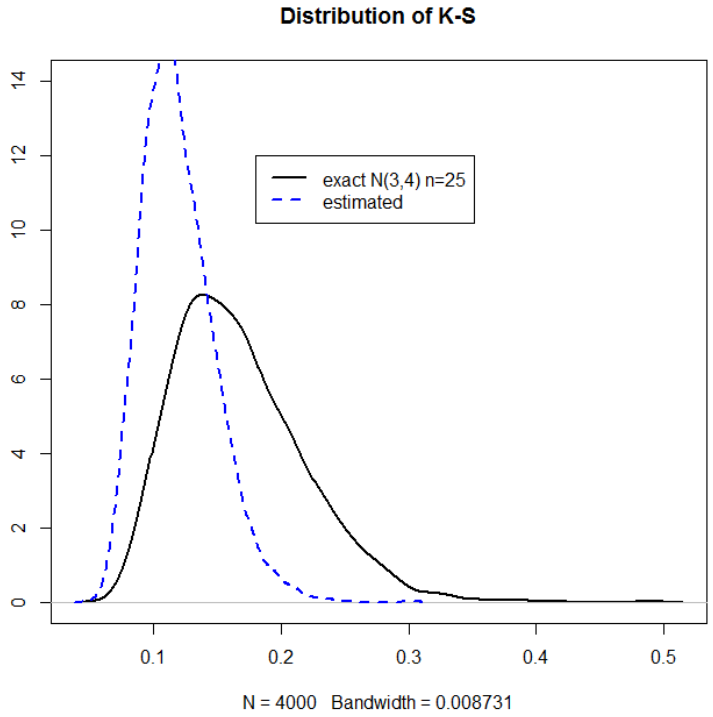
Στο Kolmogorov-Smirnov οποιαδήποτε παραδοχή για τις τιμές των παραμέτρων δεν θα πρέπει να εξαρτάται από τα δεδομένα

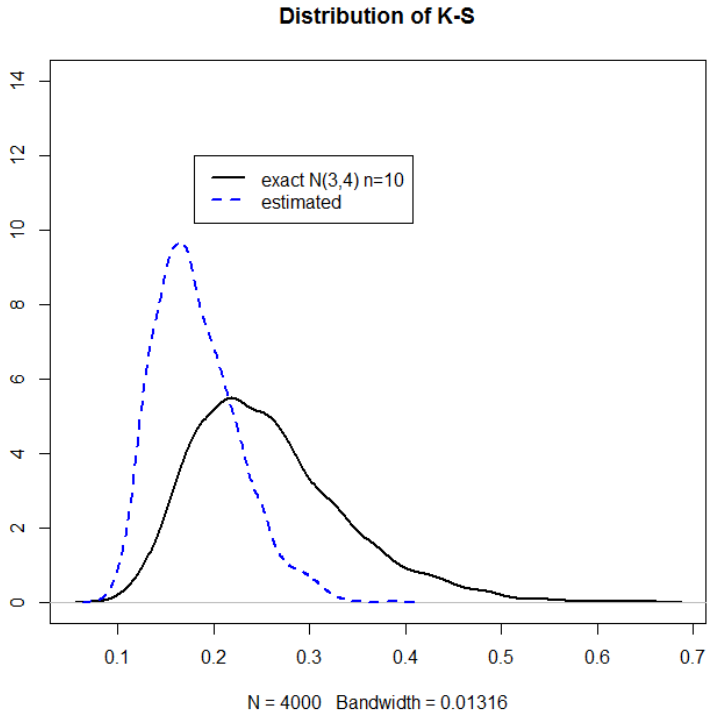




Έλεγχος για κανονικότητα

- ✓ έλεγχος του Lilliefors
- ✓ το κριτήριο των Kolmogorov-Smirnov






```
> res<-replicate(4000,{
+ x<-rnorm(10,mean=3,sd=2)
+ c(ks.test(x,"pnorm",mean=mean(x),sd=sd(x))$statistic,
+   ks.test(x,"pnorm",mean=3,sd=2)$statistic)
+ })
> plot(density(res[2,]),main="Distribution of K-S", ylab="",lwd=2,ylim=c(0,14))
> lines(density(res[1,]),lwd=2,lty=2,col="blue")
> legend(0.18,12,legend=c("exact N(3,4) n=10","estimated"),
+ lty=1:2,lwd=c(2,2),col=c(1,"blue"))
```

Σύγκριση των ελέγχων για κανονικότητα

- ✓ του Lilliefors
- ✓ των Shapiro–Wilk

```
> M<-100
> res<-replicate(M,{y<-rexp(10,1)
+ c(ks.test(y,"pnorm")$p,
+ lillie.test(y)$p,
+ shapiro.test(y)$p)
+ })
>
> round(t(res),4)
        [,1]   [,2]   [,3]
 [1,] 0.0048 0.4295 0.7290
 [2,] 0.0065 0.0432 0.0018
 [3,] 0.0029 0.3736 0.1899
 [4,] 0.0053 0.5356 0.2602
 [5,] 0.0062 0.0459 0.0277
 [6,] 0.0063 0.0000 0.0000
 [7,] 0.0039 0.0072 0.0111
 [8,] 0.0038 0.0167 0.0028
 [9,] 0.0043 0.0351 0.0310
[10,] 0.0008 0.0586 0.1532
```

Αν έχω δείγματα από Exp(1)
ποιον έλεγχο ξεγελάω ευκολότερα,
ότι αυτά προέρχονται από κανονική ;

```
> sum(t(res)[,2]>t(res)[,3])/M
[1] 0.79
> (t(res)[,2]>t(res)[,3])
```

```
 [1] FALSE
 [2]  TRUE
 [3]  TRUE
 [4]  TRUE
 [5]  TRUE
 [6]  TRUE
 [7] FALSE
 [8]  TRUE
 [9]  TRUE
[10] FALSE
```

Ο έλεγχος των Shapiro-Wilk είναι ισχυρότερος (βασιζόμενοι σε προσομοίωση), n=10

Εφαρμογές: στο t.test

Lab_4.pdf σελ. 12

The data are weight changes of humans, tabulated after administration of a drug proposed to result in weight loss. Each weight change (in kg) is the weight after minus the weight before drug administration.

0.2 -0.5 -1.3 -1.6 -0.7 0.4 -0.1 0.0 -0.6 -1.1 -1.2 -0.8

```
> x<-scan()
1: 0.2 -0.5 -1.3 -1.6 -0.7 0.4 -0.1 0.0 -0.6 -1.1 -1.2 -0.8
13:
Read 12 items
> lillie.test(x)

        Lilliefors (Kolmogorov-Smirnov)

data:  x
D = 0.12232, p-value = 0.8892

> shapiro.test(x)

        Shapiro-Wilk normality test

data:  x
W = 0.96694, p-value = 0.8763

> lillie.test(x)$p; shapiro.test(x)$p
[1] 0.8891601
[1] 0.8762924
```

```
> m0<--0.8
> t.test(x,mu=m0)

        One Sample t-test

data:  x
t = 1.0487, df = 11, p-value = 0.3168
alternative hypothesis: true mean is not
95 percent confidence interval:
 -1.0105948 -0.2060719
sample estimates:
 mean of x
-0.6083333
```

H_0: μ=-0.8

H_1: μ≠-0.8

Εφαρμογές: στο var.test και t.test

Lab_4.pdf σελ. 26-31

Data about the cost of a meal per person from a sample of 50 city restaurants

Data about the cost of a meal per person from a sample of 40 suburban restaurants.

Suburban Cost Data									
21	22	25	25	26	26	27	27	28	28
28	29	31	32	32	35	35	36	37	37
37	38	38	38	39	40	40	41	41	41
42	42	43	44	47	47	47	48	50	50

Does a meal per person cost the same in both types of restaurants?

$H_0: \sigma^2_1 = \sigma^2_2 \ (\sigma^2_1/\sigma^2_2 = 1)$

$H_1: \sigma^2_1 \neq \sigma^2_2 \ (\sigma^2_1/\sigma^2_2 \neq 1)$

```
> x<-CityCost
> y<-SuburbanCost
> n<-length(x)
> m<-length(y)
>
> lillie.test(x)$p; shapiro.test(x)$p
[1] 0.3415578
[1] 0.4942601
> lillie.test(y)$p; shapiro.test(y)$p
[1] 0.2097248
[1] 0.1820923
```

```
> a<-0.01
> d0<-1
> var.test(x,y,ratio=d0,conf.

        F test to compare two

data:  x and y
F = 2.962, num df = 49, denom
> var.test(x,y,ratio=d0,conf.
[1] 0.0006742548
```

Εφαρμογές: στο var.test και t.test

Lab_4.pdf σελ. 33-34

Bacterial Densities. The data below are the natural logarithms of bacterial counts as measured by two analysts on identical aliquots of river water. Are the analysts getting the same result?

Analyst A	1.60	1.74	1.72	1.85	1.76	1.72	1.78
Analyst B	1.72	1.75	1.55	1.67	2.05	1.51	1.70

```
> lillie.test(AnalystA-AnalystB)$p
[1] 0.5189018
> shapiro.test(AnalystA-AnalystB)$p
[1] 0.328875
> t.test(AnalystA,AnalystB,paired=T)$p.value
[1] 0.6674726
```

Εφαρμογές: στο var.test και t.test

Lab_3.pdf σελ. 10

The average amount of precipitation (rainfall) in inches for each of 70 United States (and Puerto Rico) cities.

```
> lillie.test(precip)$p; shapiro.test(precip)$p
[1] 0.03812166
[1] 0.04492529
> t.test(precip,mu=35)

        One Sample t-test

data:  precip
t = -0.069761, df = 69, p-value = 0.9446
alternative hypothesis: true mean is not equal to 35
95 percent confidence interval:
 31.61748 38.15395
sample estimates:
mean of x
 34.88571

> ks.test(precip,"pnorm",mean=35,sd=13.7)$p
[1] 0.4124419
```

